

This is a trial paper and does not necessarily reflect the content or format of the final HSC paper



Trial Higher School Certificate 2000

3/4 Unit Mathematics

**Total Time Allowed: 2 hours
(plus 5 minutes reading time)**

Instructions to Candidates:

- There are seven questions each worth ¹²~~15~~ marks.
- Each question attempted is to be returned in a separate writing booklet clearly marked *Question 1, Question 2 etc* on the cover. Each booklet must show your student number.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Standard Integrals are printed at the end of the paper.
- Even if you have not attempted a question submit a numbered blank booklet clearly showing your student number.
- In every question all necessary working is to be shown in pen except for diagrams which should be large and drawn in pencil.
- NSW Board of Studies approved calculators may be used.

Question 1 (15marks) Use a separate writing booklet

Marks

(a) Find (i) $\int \frac{x+3}{x^2+6x-7} dx$ 5

(ii) $\int_0^{\frac{\pi}{4}} \cos^2 x dx$

(b) By using the substitution $u = 1+x^2$ or otherwise find 4

$$\int \frac{4x}{\sqrt{1+x^2}} dx$$

(c) The velocity of a particle is given by $\frac{dx}{dt} = 4t - 7$ 2
Given that $x=3$ when $t=0$, find an expression for the distance x at any time t .

(d) Use the substitution $x = 3\sin\theta$ to solve $\int_0^3 x\sqrt{9-x^2} dx$ 4

Question 2 (15 marks) Use a separate writing booklet

Marks

(a) Using exact values, show that $\sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$ 3

(b) Find the general solution for x given $\cos^2 x - 3\sin x + 3 = 0$ 4

(c) Prove the identity $\frac{\cos A + \sin A}{\cos A - \sin A} - \frac{\cos A - \sin A}{\cos A + \sin A} = 2 \tan 2A$ 4

(d) $x = 2$ is a zero of the polynomial $P(x) = x^3 + x^2 + kx - 4$ 4

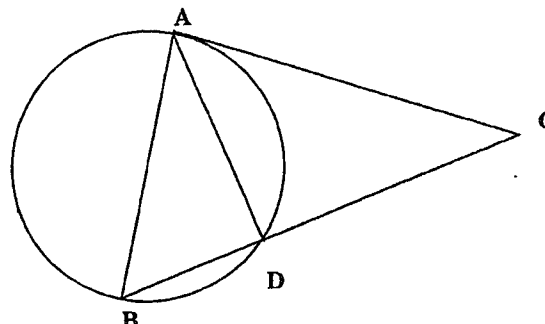
- (i) Determine the value of k
(ii) Determine all the factors of $P(x)$

Question 3 (15 marks) Use a separate writing booklet

Marks
8

- (a) In the diagram, AC is a tangent to the circle at A and angle $CAB = 90^\circ$

- (i) Show that $\triangle ABD \parallel \triangle CAD$
(ii) given that $BD=4\text{cm}$ and $CD=6\text{cm}$ calculate the length of AD
(iii) Calculate the radius of the circle.



- (b) A pump is used to inflate a spherical balloon. It is found that when the radius of the balloon is increasing at the rate of 1cm/s , the radius is 40cm .

7

- (i) At what rate is the volume of the balloon increasing when the radius is 40cm ?
(ii) Determine the rate at which the surface area of the balloon would increase when its radius is 40cm .

Question 4 (15 marks) Use a separate writing booklet

Marks

- (a) Differentiate w.r.t. x

3

$$y = \cos^{-1}(\sin x)$$

- (b) Write down the domain and range of the function

3

$$y = 2\cos^{-1} 3x$$

Draw a neat sketch of the function.

- (c) Show that $\int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{9+4x^2} = \frac{\pi}{36}$

4

- (d) (i) Show that $\cos^{-1}(-x) = \pi - \cos^{-1} x$

5

- (ii) Hence evaluate $\sin^{-1} \frac{1}{2} + \cos^{-1} \left(-\frac{\sqrt{3}}{2}\right)$

Question5 (15 marks) Use a separate writing booklet

Marks

- (a) A particle moves in a straight line so that its position x from a fixed point O at time t is given by: 10
$$x = 3\sin 2t + 4\cos 2t$$
- (i) If the motion is expressed in the form $x = r \sin(2t + \alpha)$, evaluate the constants r and α .
- (ii) Show that the motion is Simple Harmonic.
- (iii) What is the period of oscillation?
- (iv) Determine the maximum displacement from the centre of motion.
- (b) Prove by mathematical induction that 5
$$1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$$

Question6 (15 marks) Use a separate writing booklet

Marks

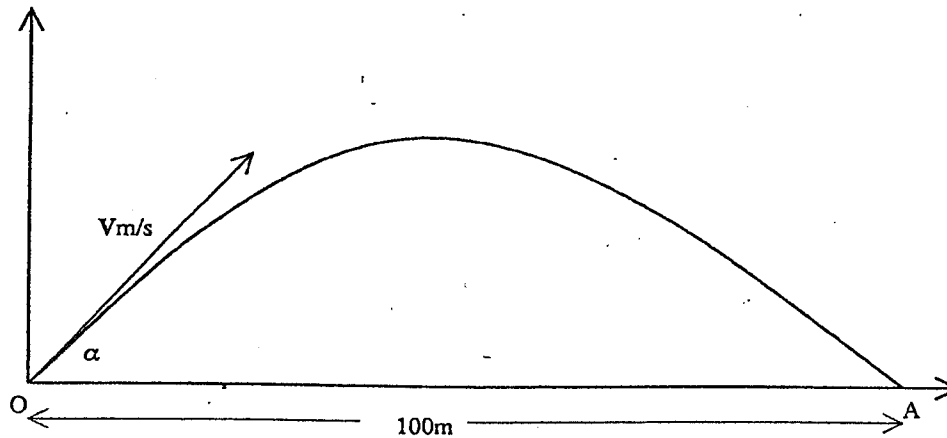
- (a) Newton's law of cooling can be represented mathematically as 7
$$\frac{dT}{dt} = -k(T - T_0)$$
 where $\frac{dT}{dt}$ is the rate of cooling, k is a constant, T is the temperature at any instant and T_0 is the room temperature.
- (i) Show that $T = T_0 + Ae^{-kt}$ is a solution to the above equation.
- (ii) A cup of tea cools from 85°C to 80°C in 1 minute. Taking room temperature as 25°C , find, to the nearest degree, the temperature of the tea after 4 minutes.
- (b) Determine the co-ordinates of the point P that divides the line joining A(-1,6) and B(4,-6) externally in the ratio 2:3 4
- (c) Find the acute angle between the lines $2x - y - 3 = 0$ and $x - 3y - 7 = 0$ 4

Question 7 (15 marks) Use a separate writing booklet

Marks

- (a) A projectile is fired from ground level at an angle α to the horizontal, with initial velocity V metres per second. The projectile returns to the ground after 5 seconds, 100m away from the point of projection. Assume that the acceleration due to gravity is 10ms^{-2} , and that the ground is horizontal.

12 [(a) + (b)]



- (i) Beginning with $\ddot{x} = 0$ and $\ddot{y} = -10$, derive equations for velocity and displacement. (i.e. for $\dot{x}, \dot{y}, x, y$) as functions of time.
- (ii) Calculate the angle of projection to the nearest minute
- (iii) Find the initial velocity in exact form.
- (iv) Find the maximum height attained by the projectile. (to the nearest metre)
- (b) At the same time that the projectile in (a) is fired, a man, 2metres tall, (unaware that the projectile has been fired) walks from A towards O. A few seconds later, he is hit on the top of his head by the projectile. Show that this accident occurs at a distance of approximately 98 metres from O.
- (c) The function $y = 4x^2 - 11x + 7$ has an approximate root at 0.73 3
Using one application of Newton's method, determine a closer approximation to the root.

$$(i) \int \frac{x+3}{x^2+6x-7} dx$$

$$= \sqrt{\frac{1}{2} \log_e (x^2+6x-7)} + C \checkmark$$

$$(ii) \int_0^{\pi/4} \cos^2 x dx$$

$$= \int_0^{\pi/4} \frac{1+\cos 2x}{2} dx \checkmark$$

$$= \left[\frac{x}{2} + \frac{\sin 2x}{4} \right]_0^{\pi/4} \checkmark$$

$$= \left[\frac{\pi}{8} + \frac{\sin 2 \times \frac{\pi}{4}}{4} \right] - (0) \checkmark$$

$$= \frac{\pi}{8} + \frac{\sin \frac{\pi}{2}}{4}$$

$$= \frac{\pi}{8} + \frac{1}{4}$$

$$= \frac{1}{8} (\pi+2) \checkmark$$

2000

(b).

$$\int \frac{4x}{\sqrt{1+x^2}} dx$$

$$= \int \frac{4x}{u^{1/2}} \cdot \frac{1}{2x} du$$

$$= 2 \int u^{-1/2} du$$

$$= 2 \cdot \frac{u^{1/2}}{1/2} + A$$

$$= 4\sqrt{u} + A$$

$$= \underline{\underline{4\sqrt{1+x^2} + A}}$$

$$u = 1+x^2$$

$$du = 2x \cdot dx$$

$$\therefore dx = \frac{1}{2x} du$$

$$(c) \frac{dx}{dt} = 4t-7$$

$$\therefore \int dx = \int (4t-7) dt + A$$

$$\therefore x = 2t^2 - 7t + A$$

$$\text{When } t=0, x=3$$

$$\therefore 3 = A$$

$$\therefore \boxed{x = 2t^2 - 7t + 3}$$

$$\int_0^{\pi/2} x \sqrt{9-x^2} dx, \quad x = 3 \sin \theta$$

$$\therefore dx = 3 \cos \theta \cdot d\theta$$

$$\text{When } x=3, \sin \theta = 1$$

$$\therefore \theta = \frac{\pi}{2}$$

$$= \int_0^{\pi/2} 3 \sin \theta \sqrt{9-9 \sin^2 \theta} \cdot 3 \cos \theta d\theta$$

$$\text{When } x=0, \sin \theta = 0$$

$$\therefore \theta = 0$$

$$= \int_0^{\pi/2} 3 \sin \theta \cdot 3 \cos \theta \cdot 3 \cos \theta d\theta$$

$$= 27 \int_0^{\pi/2} \cos^2 \theta \cdot \sin \theta d\theta$$

$$= 27 \cdot \left[\frac{\cos^3 \theta}{-3} \right]_0^{\pi/2}$$

$$= 9 [0 - 1]$$

$$= \underline{\underline{-9}}$$

(a) To show: $\sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$

$$\sin 75^\circ = \sin (45^\circ + 30^\circ)$$

$$= \sin 45^\circ \cdot \cos 30^\circ + \cos 45^\circ \cdot \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2}$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}} //$$

b. $\cos^2 x - 3 \sin x + 3 = 0$

$$1 - \sin^2 x - 3 \sin x + 3 = 0$$

$$1 - \sin^2 x - 3 \sin x + 4 = 0$$

$$1 - \sin^2 x + 3 \sin x - 4 = 0$$

$$(1 - \sin^2 x + 3 \sin x - 4) = 0$$

$$\therefore \sin x = 1, -1$$

$$\therefore x = n\pi + (-1)^n \frac{\pi}{2}$$

$$\frac{\cos A + \sin A}{\cos A - \sin A} - \frac{\cos A - \sin A}{\cos A + \sin A} = 2 \tan 2A$$

$$\text{LHS} = \frac{\cos A + \sin A}{\cos A - \sin A} - \frac{\cos A - \sin A}{\cos A + \sin A}$$

$$= \frac{(\cos A + \sin A)^2 - (\cos A - \sin A)^2}{\cos^2 A - \sin^2 A}$$

$$= \frac{\cos^2 A + 2\sin A \cos A + \sin^2 A - [\cos^2 A - 2\sin A \cos A + \sin^2 A]}{\cos 2A}$$

$$= \frac{\cancel{\cos^2 A} + 2\sin A \cos A + \cancel{\sin^2 A} - \cancel{\cos^2 A} + 2\sin A \cos A - \cancel{\sin^2 A}}{\cos 2A}$$

$$= \frac{4 \sin A \cos A}{\cos 2A}$$

$$= \frac{2 \sin 2A}{\cos 2A}$$

$$= 2 \tan 2A$$

$$= \text{RHS}$$

①

$$P(x) = x^3 + x^2 + Kx - 4$$

$$x = 2 \text{ is a Solution}$$

$$\therefore P(2) = 0$$

$$8 + 4 + 2K - 4 = 0$$

$$12 + 2K = 0$$

$$K = -4$$

$$P(x) = x^3 + x^2 - 4x - 4$$

$$\begin{array}{r} x^2 + 3x + 2 \\ x-2 \overline{) x^3 + x^2 - 4x - 4} \\ \underline{x^3 - 2x^2} \\ 3x^2 - 4x - 4 \\ \underline{3x^2 - 6x} \\ 2x - 4 \\ \underline{2x - 4} \\ 0 \end{array}$$

$$x^2 + 3x + 2 = (x+2)(x+1)$$

$$\therefore P(x) = (x-2)(x+2)(x+1)$$

Given: AC is a tangent at A
 $\angle BAC = 90^\circ$
 $BD = 4 \text{ cm}$, $DC = 6 \text{ cm}$

Prove: (i) $\triangle ABD \sim \triangle CAD$

Proof: (ii) Calculate length of radius

$\angle BAC = 90^\circ$ — data

$\therefore AB$ is the diameter

$\therefore \angle ADB = 90^\circ$ — (angle in a semicircle)

In $\triangle ABD$, $\triangle CAD$

$\angle ABD = \angle CAD$ — $\angle$ s in the alternate segment

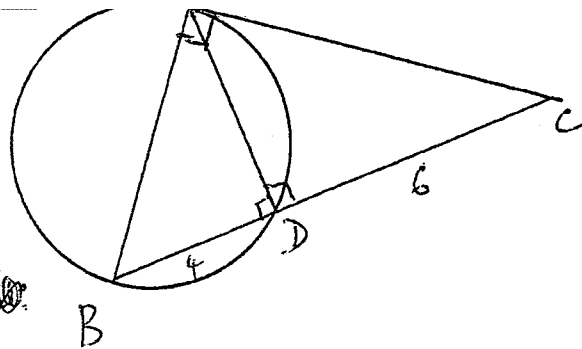
$\angle ADB = \angle ADC = 90^\circ$ — given (adj $\angle$ s on a straight line)

$\therefore \triangle ABD \sim \triangle CAD$ (A, A, $\therefore$ A)

$AC^2 = CB \cdot CD$ — (tangent² = product of intercepts of chord)

$= 10 \times 6$
 $\therefore AC = \sqrt{60}$

$AD^2 = AC^2 - CD^2$
 $= 60 - 36$
 $= 24$



$AD = 2.16 \text{ cm}$ ✓

$$AB^2 = AD^2 + BD^2$$

$$= 24 + 16$$

$$= 40$$

$$\therefore AB = 2\sqrt{10}$$

$$\therefore \text{radius} = \frac{1}{2} AB = \sqrt{10} \text{ cm}$$



(b) Given: $\frac{dr}{dt} = 1 \text{ cm/s}$ when $r = 40 \text{ cm}$.

(i)

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$V = \frac{4}{3}\pi r^3$$

$$\therefore \frac{dV}{dr} = \frac{4}{3} \times \pi \times 3r^2$$

$$= 4\pi r^2$$

$$\therefore \frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$

$\therefore$ when $r = 40 \text{ cm}$

$$\left(\frac{dV}{dt}\right)_{r=40} = 4 \times \pi \times 40^2 \times 1$$

$$= 6400\pi \text{ cm}^3/\text{s}$$

$\frac{36}{32}$

11) $\omega = 4\pi \text{ rad/s}$ ✓ $\frac{ds}{dt} = 8\pi r$ ✓

$$\frac{ds}{dt} = \frac{ds}{dr} \cdot \frac{dr}{dt}$$

$$= 8\pi r \cdot \frac{dr}{dt}$$

$$\left(\frac{ds}{dt}\right)_{r=40\text{cm}} = 8 \times \pi \times 40 \times 1$$

$$= 320\pi \text{ cm}^2/\text{s}$$

Question 12

(a) $y = \cos^{-1}(\sin x)$

$$\therefore \cos y = \sin x$$

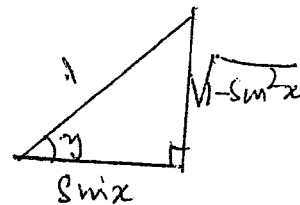
$$\therefore -\sin y \frac{dy}{dx} = \cos x$$

$$\therefore \frac{dy}{dx} = -\frac{\cos x}{\sin y}$$

$$= -\frac{\cos x}{\sqrt{1 - \sin^2 x}}$$

$$= -\frac{\cos x}{\cos x}$$

$$= -1$$



or:

$$\cos y = \sin x$$

$$\therefore x + y = 90^\circ$$

$$\therefore 1 + \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -1$$

(b) $y = 2 \cos^{-1} 3x$

$$\therefore \frac{y}{2} = \cos^{-1} 3x$$

$$D: -1 \leq 3x \leq 1$$

$$\text{i.e. } -\frac{1}{3} \leq x \leq \frac{1}{3}$$

$$R: 0 \leq \frac{y}{2} \leq \pi$$

$$\therefore 0 \leq y \leq 2\pi$$

$$\begin{aligned}
 & \int \frac{1}{9+4x^2} \\
 &= \int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{4(\frac{9}{4}+x^2)} \\
 &= \frac{1}{4} \int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{(\frac{3}{2})^2+x^2} \\
 &= \frac{1}{4} \cdot \frac{1}{\frac{3}{2}} \left[\tan^{-1} \frac{x}{\frac{3}{2}} \right]_0^{\frac{\sqrt{3}}{2}} \\
 &= \frac{1}{6} \left[\tan^{-1} \frac{2x}{3} \right]_0^{\frac{\sqrt{3}}{2}} \\
 &= \frac{1}{6} \left[\tan^{-1} \frac{1}{\sqrt{3}} \right] \\
 &= \frac{1}{6} \times \frac{\pi}{6} \\
 &= \frac{\pi}{36}
 \end{aligned}$$

$$\cos^{-1}(-x) = \pi - \cos^{-1}x$$

$$\text{Let } y = \cos^{-1}(-x)$$

$$\therefore \cos y = -x$$

$$\therefore x = -\cos y$$

$$\text{i.e. } x = \cos(\pi - y)$$

$$\therefore \cos^{-1}x = \pi - y$$

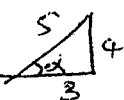
$$\therefore y = \pi - \cos^{-1}x$$

$$\text{i.e. } \cos^{-1}(-x) = \pi - \cos^{-1}x$$

$$\begin{aligned}
 \text{(ii)} \quad & \sin^{-1} \frac{1}{2} + \cos^{-1} \left(-\frac{\sqrt{3}}{2} \right) \\
 &= \sin^{-1} \frac{1}{2} + \pi - \cos^{-1} \frac{\sqrt{3}}{2} \\
 &= \frac{\pi}{6} + \pi - \frac{\pi}{6} \\
 &= \underline{\underline{\pi}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(a)} \quad x &= 3 \sin 2t + 4 \cos 2t \\
 &= 5 \left[\frac{3}{\sqrt{3^2+4^2}} \sin 2t + \frac{4}{\sqrt{3^2+4^2}} \cos 2t \right] \\
 \text{i.e. } x &= 5 \left[\frac{3}{5} \sin 2t + \frac{4}{5} \cos 2t \right] \\
 &= 5 \left[\sin 2t \cos \alpha + \cos 2t \sin \alpha \right]
 \end{aligned}$$

where $\cos \alpha = \frac{3}{5}$



$$\text{i.e. } x = 5 \sin(2t + \alpha)$$

$$\dot{x} = 10 \cos(2t + \alpha)$$

$$\ddot{x} = -20 \sin(2t + \alpha)$$

$$= -4 \times 5 \sin(2t + \alpha)$$

$$\therefore \ddot{x} = -4x$$

This is of the form $\ddot{x} = -n^2 x$

$\therefore$ The motion is Simple Harmonic.

$$\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi \text{ seconds.}$$

$$\text{Max. displacement} = \text{Amplitude} = \underline{\underline{5 \text{ cm}}}$$

(b) To show that.

$$1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$$

Let this be true for $n = k$. When $k \geq 1$

$$\therefore 1 + 4 + 7 + \dots + (3k-2) = \frac{k(3k-1)}{2} \quad \text{--- (1)}$$

Adding the next, $(k+1)^{\text{th}}$ term:

$$1 + 4 + 7 + \dots + (3k-2) + (3k+1) = \frac{k(3k-1)}{2} + (3k+1)$$

$$= \frac{k(3k-1) + 2(3k+1)}{2}$$

$$= \frac{3k^2 - k + 6k + 2}{2}$$

$$= \frac{3k^2 + 5k + 2}{2}$$

$$= \frac{(k+1)(3k+2)}{2} \quad \text{--- (2)}$$

This is of the same form as (1) when k is replaced by $k+1$.

$\therefore$ If Statement (1) is true for $n = k$, it is true for $n = k+1$ also.

When $k=1$, LHS in (1) = 1, RHS in (1) = $1 \cdot \frac{(3-1)}{2} = 1$.

$\therefore$ By the Principle of Induction: $1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$

Question No 6

$$\textcircled{a} \quad \frac{dT}{dt} = -k(T-T_0)$$

$$\therefore \frac{dT}{T-T_0} = -k dt$$

$$\therefore \int \frac{dT}{T-T_0} = -k \int dt + C$$

$$\therefore \log_e (T-T_0) = \frac{-kt + C}{-kt + C}$$

$$\therefore \frac{T-T_0}{T-T_0} = e^{-kt}$$

$$\therefore T = T_0 + Ae^{-kt}$$

Alternative method

$$T = T_0 + Ae^{-kt}$$

$$\frac{dT}{dt} = -kT$$

But from (1)

$$Ae^{-kt} = T - T_0$$

$$\therefore \frac{dT}{dt} = -k(T - T_0)$$

$$(b) \quad T = 85^\circ \text{ when } t = 0.$$

$$T = 80 \text{ when } t = 1 \text{ min}$$

$$T = ? \text{ when } t = 4 \text{ min}$$

$$T_0 = 25^\circ$$

$$T = T_0 + Ae^{-kt}$$

$$\text{i.e. } T = 25 + Ae^{-kt}$$

$$\text{when } t = 0, T = 85$$

$$85 = 25 + A$$

$$\therefore A = 60$$

When $t=1$, $T=80^\circ$

$$80 = 25 + 60 \cdot e^{-k}$$

$$\text{ie } 60e^{-k} = 55$$

$$\therefore e^{-k} = \frac{55}{60}$$

$$\text{ie } -k = \log_e \frac{55}{60}$$

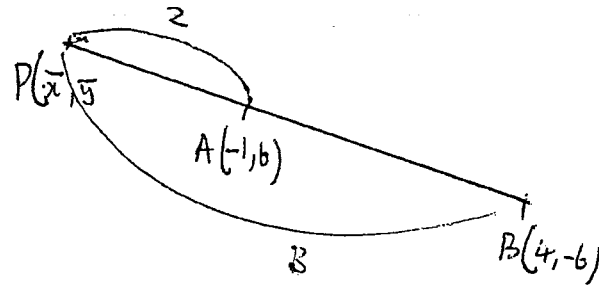
$$\therefore k = \underline{0.087011376}$$

When $t=4$ min $T=?$

$$T = 25 + 60 \cdot e^{-4k}$$

$$= 67.36^\circ$$

$$= 67^\circ$$



$$\left. \begin{array}{l} A(x_1, y_1) \\ B(x_2, y_2) \end{array} \right\} \begin{array}{l} m = 2 \\ n = 3 \end{array}$$

$$\begin{aligned} \bar{x} &= \frac{mx_2 - ny_1}{m-n} \\ &= \frac{2 \times 4 - 3 \times (-1)}{2-3} \\ &= \frac{8+3}{-1} \\ &= -11 \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{my_2 - ny_1}{m-n} \\ &= \frac{2(-6) - 3 \times 6}{2-3} \\ &= \frac{-12-18}{-1} \\ &= 30 \end{aligned}$$

$\therefore P$ is the point $(-11, 30)$.

$$\begin{aligned} \text{(c)} \quad 2x - y - 3 &= 0 \\ x - 3y - 7 &= 0 \end{aligned}$$

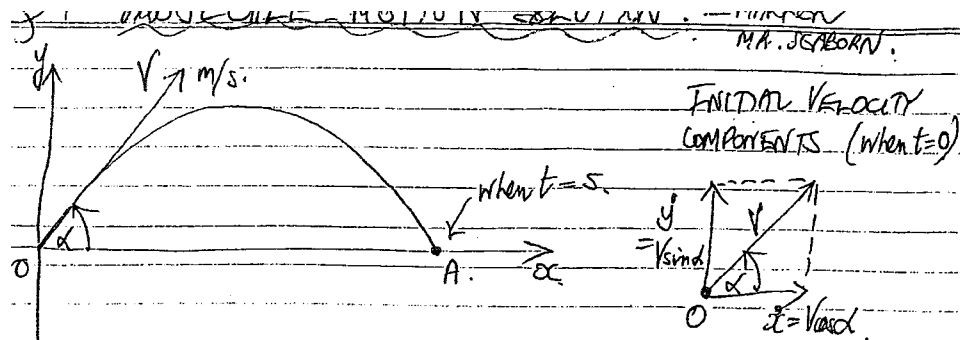
$$\begin{aligned} \text{ie } y &= 2x - 3 \quad \text{--- (1)} \\ y &= \frac{x}{3} - \frac{7}{3} \quad \text{--- (2)} \end{aligned}$$

$$\begin{aligned} m_1 &= 2 \\ m_2 &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \tan \theta &= \frac{m_1 - m_2}{1 + m_1 m_2} \\ &= \frac{2 - \frac{1}{3}}{1 + 2 \times \frac{1}{3}} = \frac{\frac{5}{3}}{\frac{5}{3}} = 1 \end{aligned}$$

$\therefore$ The Angle between the lines is 45°

(8)



rem $\ddot{x} = 0$
 $\dot{x} = \int 0 dt$
 $= C_1$

INITIALLY
 when $t=0$ $\dot{x} = V \cos \alpha \Rightarrow C_1 = V \cos \alpha$ and $\dot{y} = V \sin \alpha$ when $t=0$
 $\Rightarrow \dot{x} = V \cos \alpha$ ~~✗~~

$\ddot{y} = -10$
 $\dot{y} = \int (-10) dt$
 $= C_2 - 10t$

when $t=0$ $\dot{y} = V \sin \alpha \Rightarrow C_2 = V \sin \alpha$
 $\Rightarrow \dot{y} = V \sin \alpha - 10t$ ~~✗~~

$x = \int (V \cos \alpha) dt$
 $= (V \cos \alpha)t + C_2$
 When $t=0$ $x=0 \Rightarrow C_2=0$
 $\Rightarrow x = (V \cos \alpha)t$ — ①

$y = \int (V \sin \alpha - 10t) dt$
 $= (V \sin \alpha)t - 5t^2 + C_4$
 When $t=0$ $y=0 \Rightarrow C_4=0$
 $\Rightarrow y = (V \sin \alpha)t - 5t^2$ — ②

At A $x=100, y=0, t=5$
 $\Rightarrow 100 = (V \cos \alpha) \times 5$ and $0 = (V \sin \alpha) \times 5 - 5 \times 5^2$
 $\Rightarrow V \cos \alpha = 20$ — ③ $\Rightarrow V \sin \alpha = 25$ — ④
 $\Rightarrow \frac{25}{20} = \frac{V \sin \alpha}{V \cos \alpha} \Rightarrow \tan \alpha = \frac{5}{4} \Rightarrow \alpha = \tan^{-1}(\frac{5}{4})$

AKS $\cos \alpha = \frac{4}{\sqrt{41}}$ and $\sin \alpha = \frac{5}{\sqrt{41}}$
 $= 51.20'$ (nearest minute)

Sub in ② (or ①) to find $V \Rightarrow V = 20 \div \frac{4}{\sqrt{41}} = 5\sqrt{41} \text{ m/s}$
 1 MARK

(v) Particle is at maximum height when $\dot{y} = 0$

$\Rightarrow 0 = V \sin \alpha - 10t$
 $= 25 - 10t$

$\Rightarrow t = 2.5 \text{ s}$

To find maximum height $y_{\max} = (V \sin \alpha)t - 5t^2$ when $t=2.5$

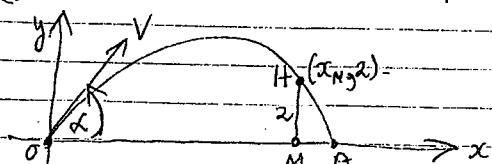
1 MARK

$= 25 \times 2.5 - 5 \times 2.5^2$

$= 31.25 \text{ m (exactly)}$

To the nearest metre the maximum height is 31 m.

(b) When the man is hit (at point H on the trajectory, say)



METHOD 1

$y = (V \sin \alpha)t - 5t^2$

$2 = 25t - 5t^2$

$5t^2 - 25t + 2 = 0$

$t = \frac{25 \pm \sqrt{625 - 40}}{10}$

$= \frac{25 \pm \sqrt{585}}{10}$

$= 0.08 \text{ s or } 4.92 \text{ s (2d)}$

"A few seconds later" means he is hit when $t \approx 4.92 = \frac{25 + \sqrt{585}}{10}$

$x = (V \cos \alpha)t$

$x_m = 20 \times \frac{25 + \sqrt{585}}{10}$

$= 50 + 2\sqrt{585}$

$= 98.3735 \text{ m (4dp)}$

$\approx 98 \text{ m}$ ~~✗~~

$\Rightarrow$ The man is hit approximately 98 m from O.

3 MARKS

METHOD 2 Equation of projectile is $y = (\tan \alpha)x - \frac{g \sec^2 \alpha}{2v^2} x^2$

$\therefore 80y = 100x - x^2$

At H: $160 = 100x - x^2$

$x^2 - 100x + 160 = 0$

$\Rightarrow x = \frac{100 \pm \sqrt{10000 - 6400}}{2}$

$= 50 \pm \sqrt{2340}$

$\approx 50 \pm 48.3735 \approx 98 \text{ m}$ since projectile hits man after a few seconds, not right away

$$b.) \quad f(x) = 4x^2 - 11x + 7$$

$$f'(x) = 8x - 11$$

$$f(0.73) = 1.016$$

$$f'(0.73) = -5.16$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\therefore = 0.73 - \frac{1.016}{-5.16}$$

$$= 0.73 + \frac{1.016}{5.16}$$

$$= 0.9435 \text{ (4 d.p.)}$$